

Deriving the Multi-Dose Concentration Formula

from a first-order differential equation
via an infinite geometric series

Beginning with the elimination law $c'(t) = k_1 c(t)$, we construct the single-dose profile, superpose n doses to obtain a finite geometric series, and pass to the limit $n \rightarrow \infty$. The steady-state peak and trough then follow by evaluating one formula at two points, and the finite- n expression reproduces the MANYDOSECONCENTRATION function exactly.

Roadmap

- Step 1.** Solve the differential equation and reparameterise by half-life
- Step 2.** Build the single-dose profile and the tail of dose k
- Step 3.** Superpose n doses: a finite geometric series
- Step 4.** Let $n \rightarrow \infty$: the series converges
- Step 5.** Evaluate at $\tau = 0$ and $\tau = P - x$: peak and trough
- Step 6.** Recover the MANYDOSECONCENTRATION function and verify against the notebooks
- Step 7.** Take the continuous limit: dosing frequency $D \rightarrow \infty$

Symbols: t time (h) · h elimination half-life (h) · x absorption duration (h) · D number of doses per P hours · P dosing period (h, default 24) · τ time since the current dose finished absorbing.

Step 1. Solve the Differential Equation

A drug is cleared at a rate proportional to the amount present. This first-order law is the entire physical input to the model:

$$c'(t) = k_1 c(t), \quad k_1 < 0. \quad (1)$$

Separating variables gives the general solution

$$c(t) = k_2 e^{k_1 t}. \quad (2)$$

Reparameterising by half-life

The constant k_1 is not measured clinically; the *half-life* h is. Imposing $c(t + h) = \frac{1}{2} c(t)$ on (2),

$$e^{k_1 h} = \frac{1}{2} \iff k_1 h = -\ln 2 \iff k_1 = -\frac{\ln 2}{h}. \quad (3)$$

Substituting (3) into (2) and using $e^{-(\ln 2)t/h} = 2^{-t/h}$ converts the solution to base two:

$$c(t) = k_2 \cdot 2^{-t/h} \quad (4)$$

Why base two

In the form $2^{-t/h}$ the half-life is read off directly: one half-life ($t = h$) gives $2^{-1} = \frac{1}{2}$, two give $2^{-2} = \frac{1}{4}$, and m give 2^{-m} . Every subsequent formula inherits this base, which is why powers of 2 rather than e appear throughout.

Step 2. The Single-Dose Profile

An oral dose is not absorbed instantaneously. Over an absorption window of length x the drug enters the blood at a constant rate; afterwards (4) governs its decay. Normalising so that one dose reaches unit concentration at $t = x$:

$$c_{\text{single}}(t) = \begin{cases} \frac{t}{x}, & 0 < t \leq x \quad (\text{absorption}), \\ 2^{-(t-x)/h}, & t > x \quad (\text{elimination}). \end{cases} \quad (5)$$

Both boundary conditions hold by construction: $c_{\text{single}}(0) = 0$ and $c_{\text{single}}(x) = 1$.

The tail left by dose k

Doses are given every P hours, so dose k is administered at time $(k-1)P$ and finishes absorbing at

$$t_k = (k-1)P + x. \quad (6)$$

For any $t > t_k$ its residual contribution is the elimination branch of (5) translated to start at t_k :

$$c_k(t) = 2^{-(t-t_k)/h} = 2^{-(t-x-(k-1)P)/h}. \quad (7)$$

Each past dose therefore leaves its own exponential tail, and successive tails are offset from one another by exactly one period P .

Step 3. Superposing n Doses

Equation (1) is linear, so contributions add. After n doses, at a time t lying in the elimination phase of the current cycle,

$$C_n(t) = \sum_{k=1}^n c_k(t) = \sum_{k=1}^n 2^{-(t-x-(k-1)P)/h}. \quad (8)$$

This sum can be factored in two equivalent ways. Both are useful, and the distinction between them matters.

Form A — anchored at the first dose

Splitting the exponent in (8) and setting $j = k - 1$,

$$C_n(t) = 2^{-(t-x)/h} \sum_{j=0}^{n-1} (2^{P/h})^j = 2^{-(t-x)/h} \cdot \frac{2^{nP/h} - 1}{2^{P/h} - 1}. \quad (9)$$

Here the ratio is $2^{P/h} > 1$: measured from the *first* dose, later terms are larger. Form A is the expression implemented in the notebooks.

Form B — anchored at the most recent dose

Let τ denote elapsed time since the *current* dose finished absorbing,

$$\tau = t - t_n = t - x - (n - 1)P, \quad 0 \leq \tau \leq P - x. \quad (10)$$

Re-indexing (8) by $i = n - k$ (cycles into the past) gives

$$C_n(\tau) = \sum_{i=0}^{n-1} 2^{-(\tau+iP)/h} = 2^{-\tau/h} \sum_{i=0}^{n-1} r^i, \quad r := 2^{-P/h} \in (0, 1). \quad (11)$$

Now the ratio is $r < 1$, and the standard partial sum applies:

$$\sum_{i=0}^{n-1} r^i = \frac{1 - r^n}{1 - r} \quad \Rightarrow \quad \boxed{C_n(\tau) = 2^{-\tau/h} \cdot \frac{1 - 2^{-nP/h}}{1 - 2^{-P/h}}} \quad (12)$$

The two forms are the same

Substituting $t = (n - 1)P + x + \tau$ into (9) and multiplying through by $2^{P/h}/2^{P/h}$ recovers (12) identically:

$$2^{-(t-x)/h} \frac{2^{nP/h} - 1}{2^{P/h} - 1} = 2^{-\tau/h} \frac{2^{P/h} - 2^{-(n-1)P/h}}{2^{P/h} - 1} = 2^{-\tau/h} \frac{1 - 2^{-nP/h}}{1 - 2^{-P/h}}.$$

Only Form B has a finite limit as $n \rightarrow \infty$, because it holds the observation point τ fixed *within* the current cycle rather than letting t run away with n . Mixing the prefactor of one form with the ratio of the other is the one algebraic trap in this derivation.

Step 4. The Limit $n \rightarrow \infty$

Since $0 < r < 1$ we have $r^n = 2^{-nP/h} \rightarrow 0$, so (12) converges:

$$C_\infty(\tau) = \lim_{n \rightarrow \infty} C_n(\tau) = \frac{2^{-\tau/h}}{1 - 2^{-P/h}} = \frac{2^{(P-\tau)/h}}{2^{P/h} - 1} \quad (13)$$

Written out, the limit is a plain geometric sum of the tails of all past doses:

$$\begin{aligned} C_\infty(\tau) &= 2^{-\tau/h} + 2^{-(\tau+P)/h} + 2^{-(\tau+2P)/h} + \dots \\ &= 2^{-\tau/h} (1 + 2^{-P/h} + 2^{-2P/h} + \dots) \\ &= \frac{2^{-\tau/h}}{1 - 2^{-P/h}}. \end{aligned} \quad (14)$$

The accumulation multiplier

The factor $(1 - 2^{-P/h})^{-1} = 2^{P/h} / (2^{P/h} - 1)$ measures how much steady-state dosing amplifies a single dose.

- $h \gg P$ (slow clearance): $2^{P/h} \approx 1 + \frac{P \ln 2}{h}$, so the multiplier is $\approx \frac{h}{P \ln 2} \gg 1$ — strong accumulation.
- $h \ll P$ (fast clearance): $2^{P/h} \gg 1$ and the multiplier tends to 1 — each dose has essentially cleared before the next.

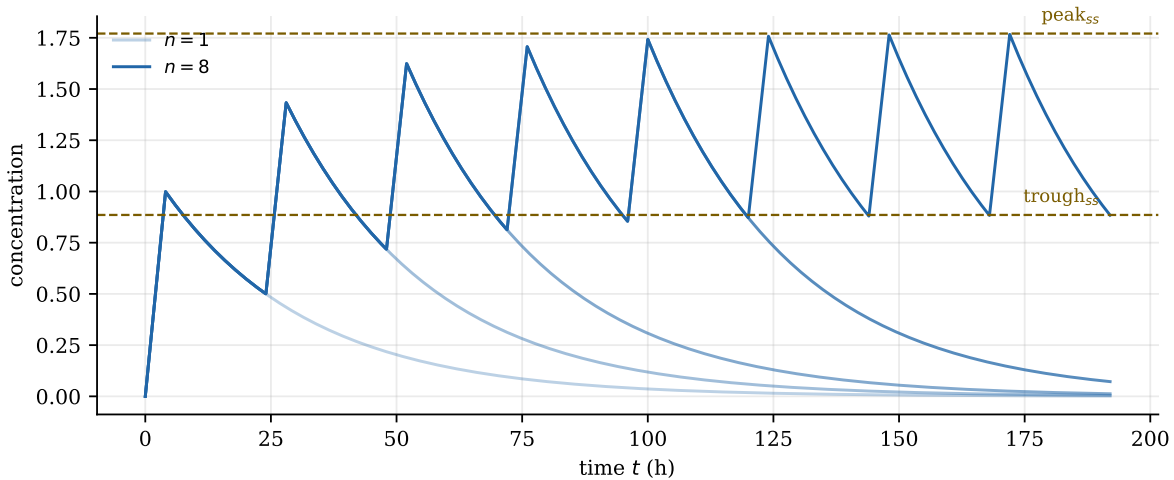


Figure 1: Successive partial sums C_n for $x = 4$ h, $h = 20$ h, $P = 24$ h. Each curve adds one dose; the profile converges to a periodic steady state bounded by the horizontal lines derived in Step 5.

Step 5. Peak and Trough

Formula (13) decreases monotonically in τ , so the extremes of a steady-state cycle occur at the two endpoints of $\tau \in [0, P - x]$.

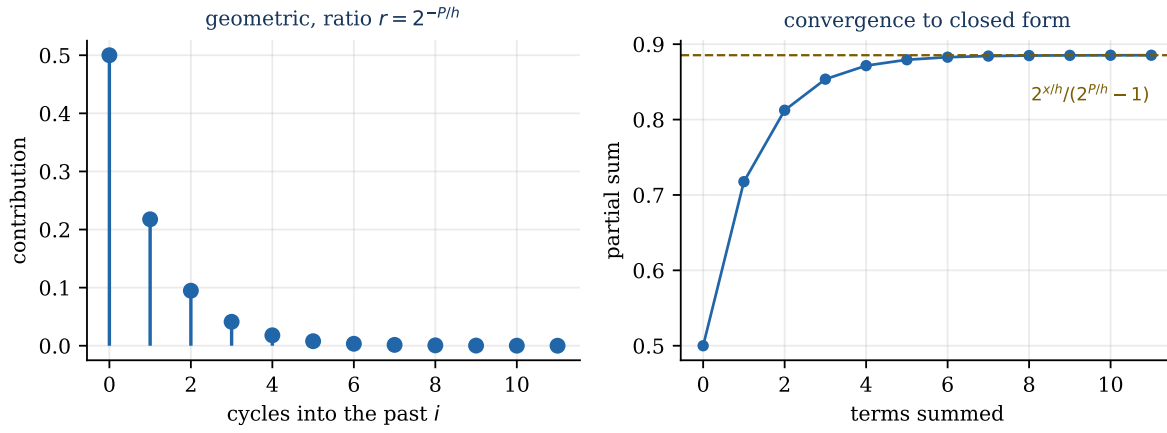


Figure 2: Left: the contributions entering the trough form a geometric sequence with ratio $r = 2^{-P/h}$. Right: their partial sums converge to the closed form $2^{x/h} / (2^{P/h} - 1)$.

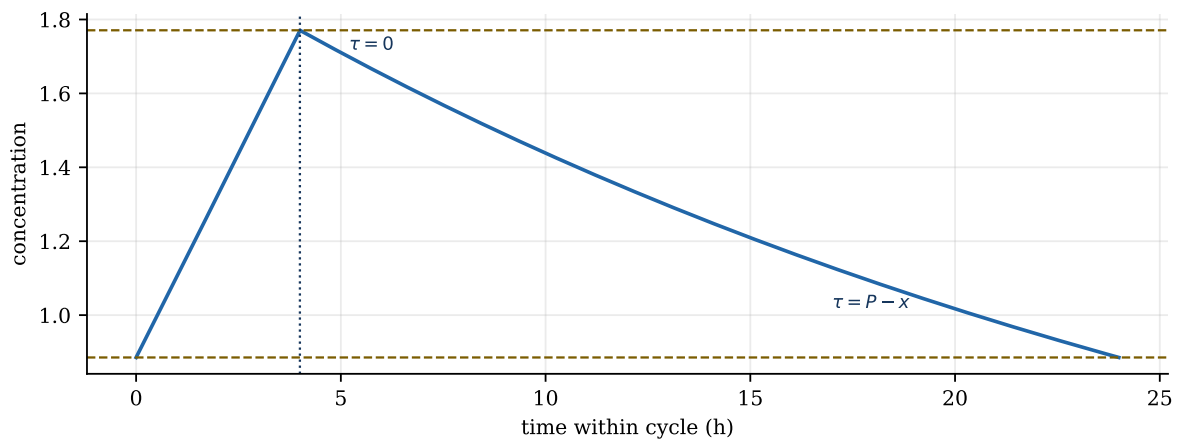


Figure 3: A single steady-state cycle. The maximum occurs the instant absorption ends ($\tau = 0$) and the minimum the instant before the next dose ($\tau = P - x$).

Peak: $\tau = 0$

At $\tau = 0$ the current dose has just reached unit concentration and all earlier tails are at their largest:

$$\text{peak}_{\text{ss}} = C_{\infty}(0) = \frac{1}{1 - 2^{-P/h}} = \frac{2^{P/h}}{2^{P/h} - 1} = 1 + \frac{1}{2^{P/h} - 1}. \quad (15)$$

The final form separates the two physical contributions: the 1 is the dose just absorbed, and $(2^{P/h} - 1)^{-1}$ is everything left over from all previous doses.

Trough: $\tau = P - x$

The minimum falls the instant before the next dose, i.e. $\tau = P - x$:

$$\text{trough}_{\text{ss}} = C_{\infty}(P - x) = \frac{2^{-(P-x)/h}}{1 - 2^{-P/h}} = \frac{2^{x/h}}{2^{P/h} - 1}, \quad (16)$$

the last step following on multiplying numerator and denominator by $2^{P/h}$. No approximation is involved: the $2^{x/h}$ arises purely from the offset between the end of absorption and the next dose.

Fluctuation index

Dividing (15) by (16), the shared denominator cancels and the ratio collapses to a single exponential:

$$\boxed{\frac{\text{peak}_{\text{ss}}}{\text{trough}_{\text{ss}}} = \frac{2^{P/h}}{2^{x/h}} = 2^{(P-x)/h}} \quad (17)$$

The swing between peak and trough depends only on how many half-lives fit into the gap $P - x$ between the end of absorption and the next dose. Extending the release window (larger x) or shortening the interval (smaller P) flattens the profile; a short half-life sharpens it.

Steady-state results

$$\text{peak}_{\text{ss}} = \frac{2^{P/h}}{2^{P/h} - 1} = 1 + \frac{1}{2^{P/h} - 1}, \quad \text{trough}_{\text{ss}} = \frac{2^{x/h}}{2^{P/h} - 1}, \quad \frac{\text{peak}_{\text{ss}}}{\text{trough}_{\text{ss}}} = 2^{(P-x)/h}.$$

Agreement with the notebook definitions

The notebooks define, with $P = 24$,

$$\text{troughmin}[x, h] = \frac{2^{x/h}}{2^{24/h} - 1}, \quad (18)$$

$$\text{peakmax}[x, h] = \frac{1}{2^{24/h} - 1} + \frac{(12 + x - 12[x/24])[x/24]}{x}. \quad (19)$$

Equation (18) is exactly (16). For (19), note that whenever $0 < x \leq 24$ we have $[x/24] = 0$, so the second term reduces to

$$\frac{(12 + x - 12) \cdot 1}{x} = \frac{x}{x} = 1,$$

and (19) becomes $1 + (2^{24/h} - 1)^{-1}$, precisely (15). The ceiling expression is therefore not an extra physical effect but the generalisation that keeps the formula valid when the absorption window spans more than one dosing period, $x > P$.

Step 6. Recovering the MANYDOSECONCENTRATION Function

For finitely many doses the notebooks evaluate Form A, (9), using ceilings to identify the current cycle. With $P = 24$ set

$$n = \lceil t/24 \rceil, \quad m = \lceil (t - x)/24 \rceil, \quad (20)$$

so that n counts doses administered and m counts those that have finished absorbing. The complete profile is

$$C(t; x, h) = \begin{cases} \frac{(12 + t - 12n)n}{x}, & 0 < t \leq x, \\ \underbrace{2^{(x-t)/h} \frac{2^{24m/h} - 1}{2^{24/h} - 1}}_{\text{Form A, eq. (9)}} - \underbrace{\frac{(n - m)(-12 - t + 12n + 12m)}{x}}_{\text{absorption-ramp correction}}, & t > x. \end{cases} \quad (21)$$

Where the series sits in the code

The first bracket of the lower branch is (9) verbatim, with nP replaced by $24m$: it is the finite geometric series summing the tails of the m doses that have completed absorption. Letting $m \rightarrow \infty$ with τ fixed returns (13), the steady state. The second bracket subtracts the portion of the linear absorption ramps already accounted for, and vanishes whenever $n = m$ — that is, whenever no dose is mid-absorption.

Step 7. Continuous IV Dosing

D doses per P hours

Everything so far assumed one dose per period, $D = 1$, with $P = 24$. Nothing in the derivation used those values. For D doses spread evenly over P hours the inter-dose interval is

$$\Delta := \frac{P}{D}, \quad (22)$$

and Steps 1–5 go through verbatim with P replaced by Δ throughout. Each dose now carries strength $1/D$, so that the total delivered per P hours is unchanged. The geometric ratio of (11) becomes $r = 2^{-\Delta/h}$ and the steady state (13) reads

$$C_{\infty}(\tau) = \frac{1}{D} \cdot \frac{2^{-\tau/h}}{1 - 2^{-\Delta/h}}, \quad 0 \leq \tau \leq \Delta - x. \quad (23)$$

Peak, trough and fluctuation follow by the same substitution:

$$\text{peak}_{ss} = \frac{1}{D} \cdot \frac{2^{\Delta/h}}{2^{\Delta/h} - 1}, \quad (24)$$

$$\text{trough}_{ss} = \frac{1}{D} \cdot \frac{2^{x/h}}{2^{\Delta/h} - 1}, \quad (25)$$

$$\frac{\text{peak}_{ss}}{\text{trough}_{ss}} = 2^{(\Delta-x)/h}. \quad (26)$$

Setting $D = 1$ recovers (15), (16) and (17) exactly. Note that the fluctuation index (26) shrinks toward 1 as D grows: splitting the same daily amount into more frequent doses flattens the profile without changing the average.

The continuous limit $D \rightarrow \infty$

Let $D \rightarrow \infty$ with P held fixed, so $\Delta = P/D \rightarrow 0$, and let $x \rightarrow 0$ so that absorption is instantaneous. Physically this is a constant intravenous infusion delivering one unit of drug every P hours, i.e. at rate

$$\lambda := \frac{1}{P} \quad (\text{units per hour}). \quad (27)$$

Take the trough (25) with $x \rightarrow 0$:

$$\text{trough}_{\text{ss}} \xrightarrow{x \rightarrow 0} \frac{1}{D} \cdot \frac{1}{2^{\Delta/h} - 1} = \frac{1}{D(2^{P/(Dh)} - 1)}. \quad (28)$$

The denominator is the only place D appears. Writing $2^u = e^{u \ln 2}$ and expanding for small $u = P/(Dh)$,

$$2^{P/(Dh)} - 1 = e^{(P \ln 2)/(Dh)} - 1 = \frac{P \ln 2}{Dh} + O\left(\frac{1}{D^2}\right). \quad (29)$$

Substituting (29) into (28), the factor D cancels:

$$C_{\infty} = \lim_{D \rightarrow \infty} \frac{1}{D \left(\frac{P \ln 2}{Dh} + O(D^{-2}) \right)} = \frac{h}{P \ln 2}. \quad (30)$$

Since $\lambda = 1/P$ by (27),

$$C_{\infty}^{\text{IV}} = \frac{\lambda h}{\ln 2} = \frac{\lambda}{k}, \quad k = \frac{\ln 2}{h} \quad (31)$$

which is the classical infusion result: steady-state concentration equals infusion rate divided by elimination rate constant. The same limit is obtained by holding $D = 1$ and letting $P \rightarrow 0$, since only the ratio $\lambda = 1/P$ survives.

Consistency check: the fluctuation vanishes

From (26) with $x \rightarrow 0$,

$$\frac{\text{peak}_{\text{ss}}}{\text{trough}_{\text{ss}}} = 2^{\Delta/h} = 2^{P/(Dh)} \xrightarrow{D \rightarrow \infty} 2^0 = 1. \quad (32)$$

Peak and trough coalesce, as they must: a continuous infusion has no peaks or troughs at all, only the flat plateau (31).

Continuous dosing as a limit of the series

The infinite geometric series of Step 4 sums the tails of infinitely many *past* doses at a fixed spacing. The continuous limit sends the spacing itself to zero while shrinking each dose in proportion, so the sum becomes an integral and the sawtooth becomes a plateau. The $1/D$ dose scaling is what makes the limit finite: without it, (28) would diverge like D . The result depends on nothing but the half-life h and the infusion rate λ — the absorption window x and the sampling time τ both drop out.

Summary

Step	Result	Mathematics
1	Exponential solution	$c' = k_1 c \Rightarrow c(t) = k_2 2^{-t/h}$
2	Single-dose profile	ramp t/x on $(0, x]$, then $2^{-(t-x)/h}$
3	Superposition of n doses	$C_n(\tau) = 2^{-\tau/h} (1 - r^n) / (1 - r)$, $r = 2^{-P/h}$
4	Convergence as $n \rightarrow \infty$	$C_\infty(\tau) = 2^{-\tau/h} / (1 - 2^{-P/h})$
5	Peak, trough, fluctuation	$2^{P/h} / (2^{P/h} - 1)$, $2^{x/h} / (2^{P/h} - 1)$, $2^{(P-x)/h}$
6	Finite- n form is MANYDOSECONCENTRATION	ceilings $n = \lceil t/24 \rceil$, $m = \lceil (t-x)/24 \rceil$
7	D doses per P hours; continuous IV limit	$\Delta = P/D$; $C_\infty^{\text{IV}} = \lambda h / \ln 2 = \lambda / k$ as $D \rightarrow \infty$, $x \rightarrow 0$

The derivation reduces to one observation: a sum of exponential tails offset by equal time steps *is* a geometric series. Once the sum is anchored at the most recent dose — so that the observation point stays inside the current cycle — the closed form, its limit, and both clinical extremes follow in a few lines of algebra.

Source: *drugs from scratch* Mathematica notebooks, Levels 1–8 (including the Cymbalta application), and the Python translation `ManyDoseConcentrationdrugs2.ipynb`.